

Origins and Evolution of Language

Class 2: Language evolution in the lab (plus the stuff I didn't cover in class 1!)

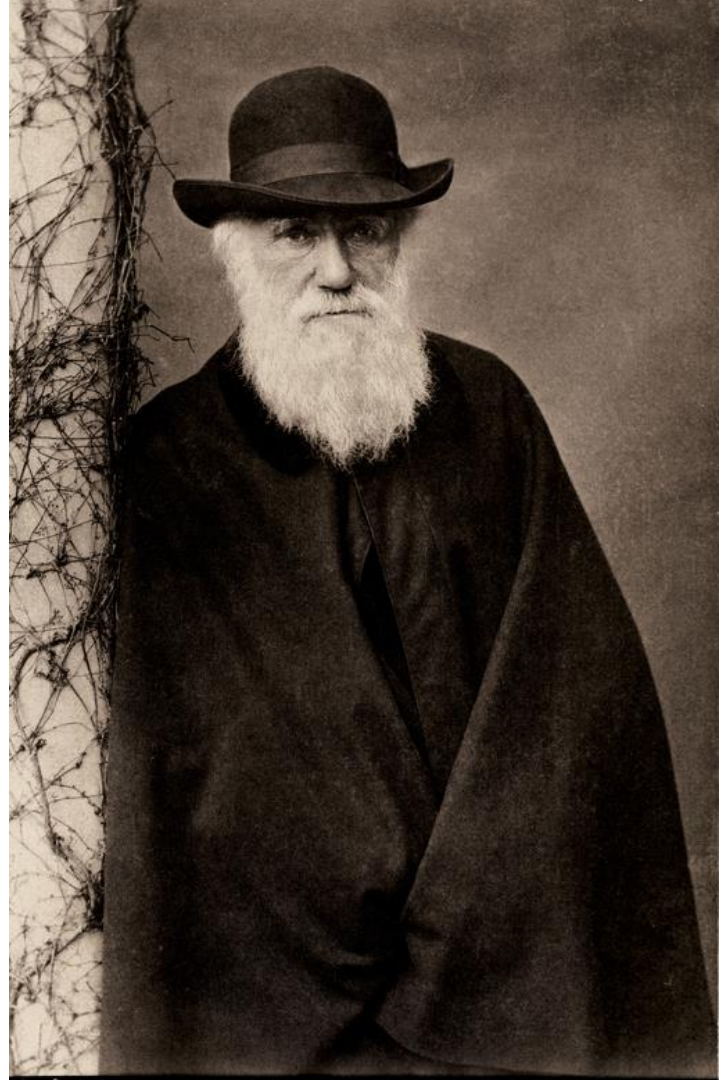
Kenny Smith

kenny.smith@ed.ac.uk

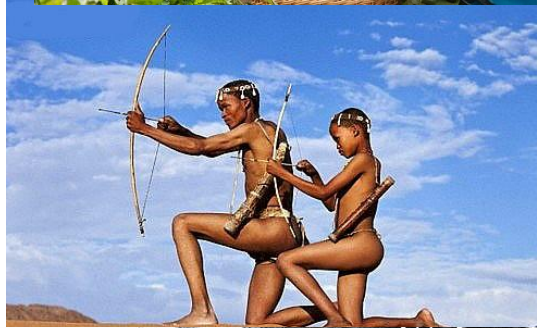
Language as an adaptation

Language as an adaptation?

- Language is just like any other **adaptive** feature of an organism's biology
- It's an **innate** feature of the human mind
- It **evolved by natural selection** under pressure for communication



A tool for the communication of knowledge and internal states



The core of the argument (from Pinker & Bloom, 1990)



“We will suggest that language shows signs of design for the communication of propositional structures over a serial channel.

The propositions in [human thought] are relational structures whose symbols pertain to people, objects, and events, the categories they belong to, their distribution in space and time, and their causal relations to one another ... The causal relations governing the behavior of other people are understood as involving their beliefs and desires, which can be reconsidered as relations between an individual and the proposition that represents the content of that belief or desire”

Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences*, 13, 707-784.
Lengthy quote is from p. 712-713

The core of the argument (from Pinker & Bloom, 1990)



“We would want to be able to

- refer to individuals and classes,*
- to distinguish among basic ontological categories (things, events, places, times, manners, and so on),*
- to talk about events and states, distinguishing the participants in the event or state according to role (agents, patients, goals),*
- to talk about the intentional states of ourselves and others*
- to express distinctions of truth value, modality (necessity, possibility, probability, factivity)*

Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences*, 13, 707-784.
Lengthy quote is from p. 712-713

The core of the argument (from Pinker & Bloom, 1990)



- *to comment on the time of an event or state, including both its distribution over time (continuous, iterative, punctate) and its overall time of occurrence*
- *to encode an unlimited number of predicates, arguments, and propositions*
- *to be able to use the same propositional content within different speech acts; for instance, as a question, a statement, or a command.*
- *to focus or to put into the background different parts of a proposition, so as to tie the speech act into its context of previously conveyed information and patterns of knowledge of the listener. ”*

The core of the argument (from Pinker & Bloom, 1990)



“All we have argued is that human language, like other specialized biological systems, evolved by natural selection. Our conclusion is based on two facts that we would think would be entirely uncontroversial: Language shows signs of complex design for the communication of propositional structures, and the only explanation for the origin of organs with complex design is the process of natural selection.” (p. 726)

Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences*, 13, 707-784.

Questions from the “ask me anything” box on the pre-lecture quiz

“I wonder, could these same pressures also lead to recursive structures like centre-embedding? Or would recursion require more specialized cognitive mechanisms?”

“when we talk about heritable variation in fitness, how do we differentiate the role of language from other cognitive factors, like working memory that is inherent in our brain?”

Questions from the “ask me anything” box on the pre-lecture quiz

“I am curious how external factors are accounted for with regards to the evolution of language. For example, there is evidence that dietary changes resulted in phonological changes. Considering this change is affected by technological developments, i.e. cultural evolution, how does this interact with the evolution of language itself? Is it a negligible effect in the long term?

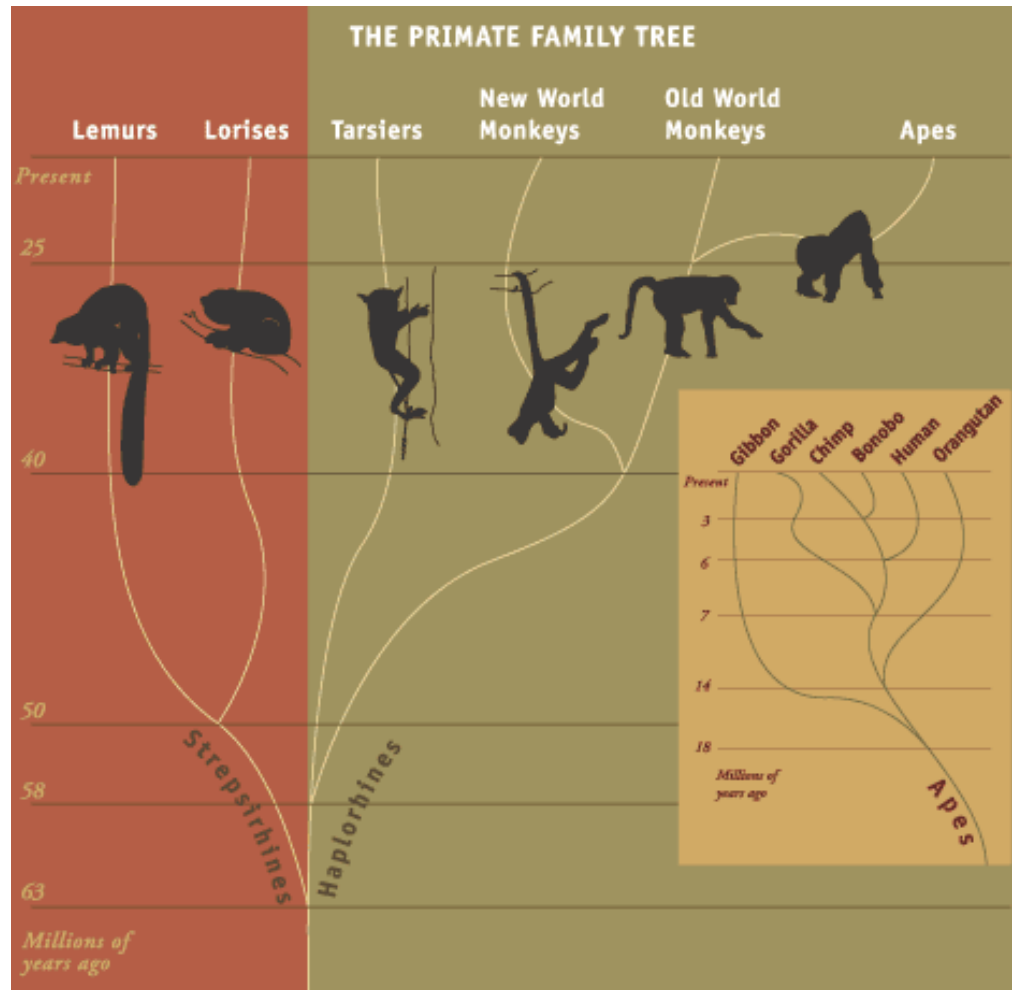
Moreover, considering the fact that linguistic evolution changes at a much higher rate than language [?] evolution, what function does this speed serve? Does, for example, an increase of available knowledge affect language change?”

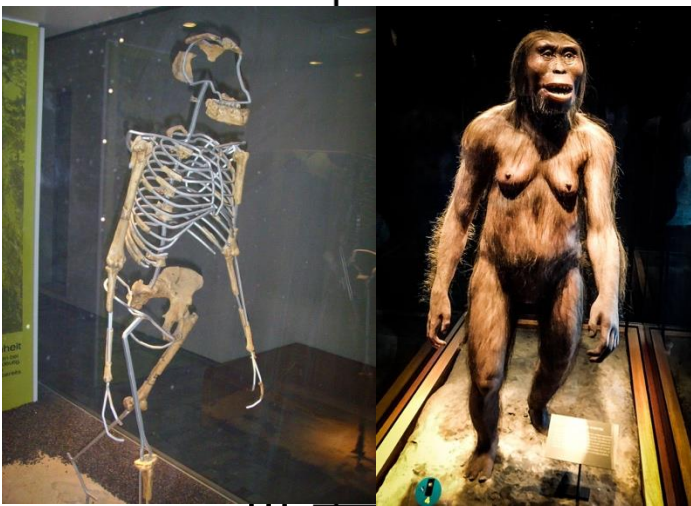
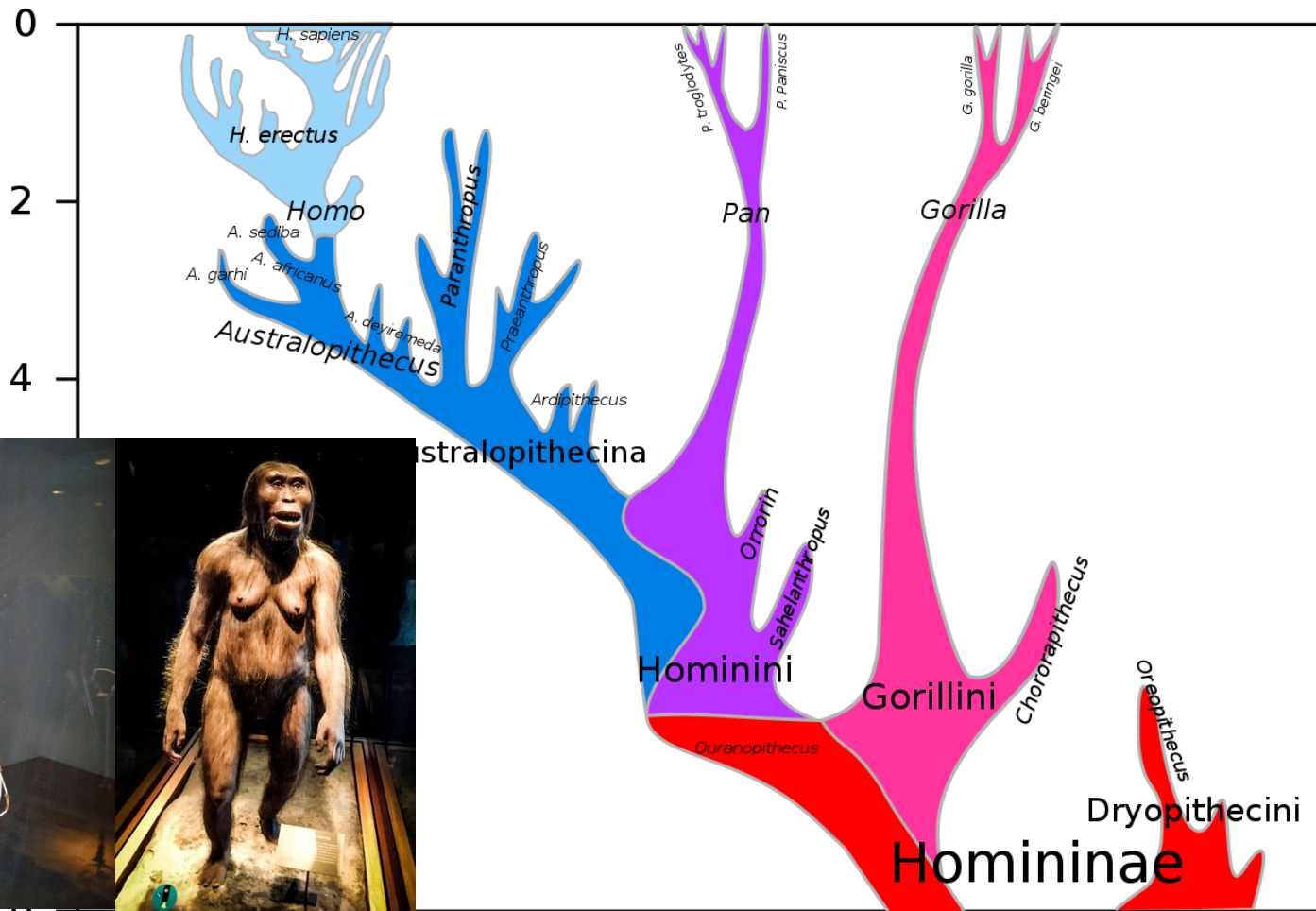
The core of the argument (from Pinker & Bloom, 1990)



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Basic of human evolution
(as an excuse to talk about **evidence**)



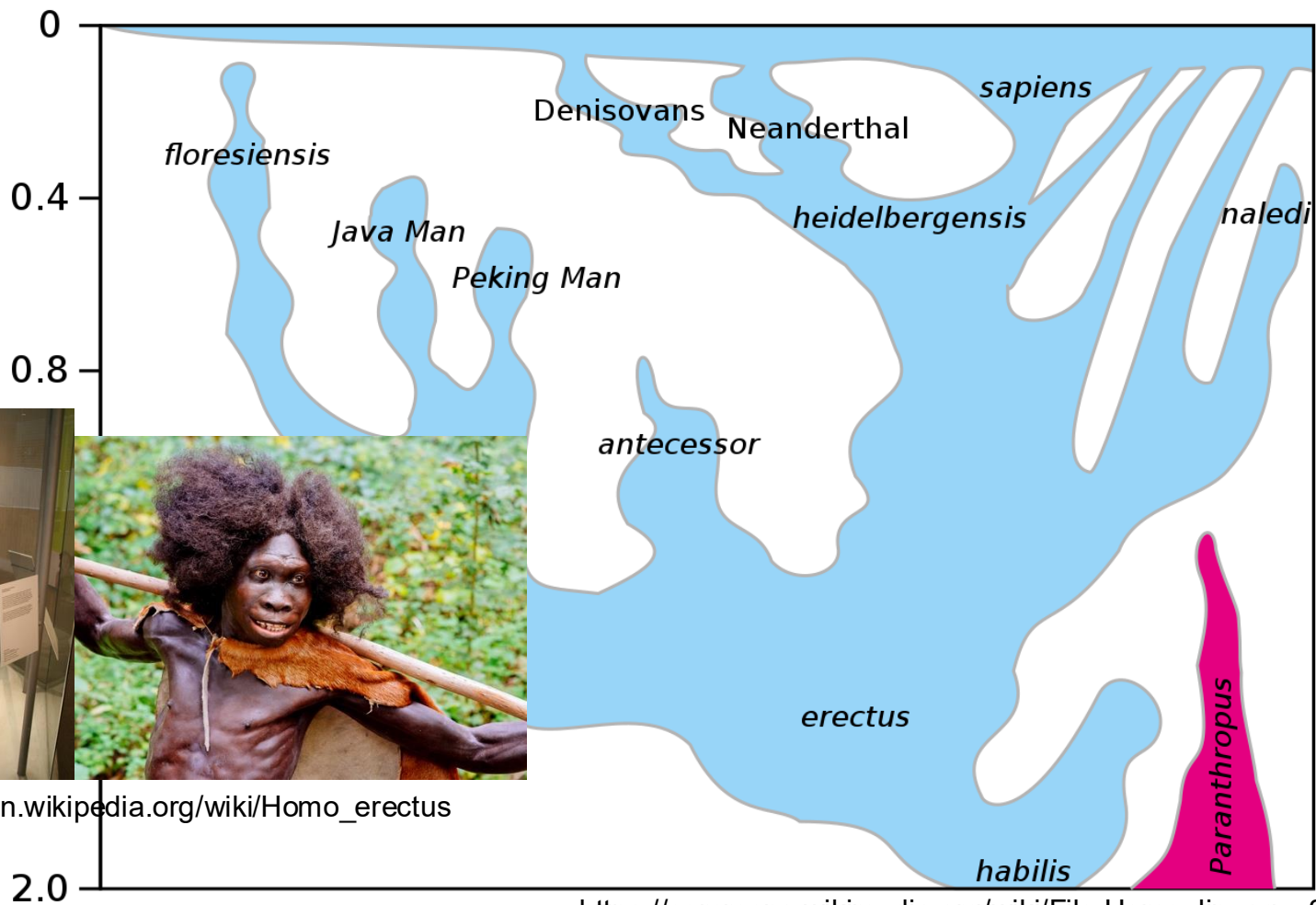


https://en.wikipedia.org/wiki/Lucy_%28Australopithecus%29

https://commons.wikimedia.org/wiki/File:Hominini_lineage.svg

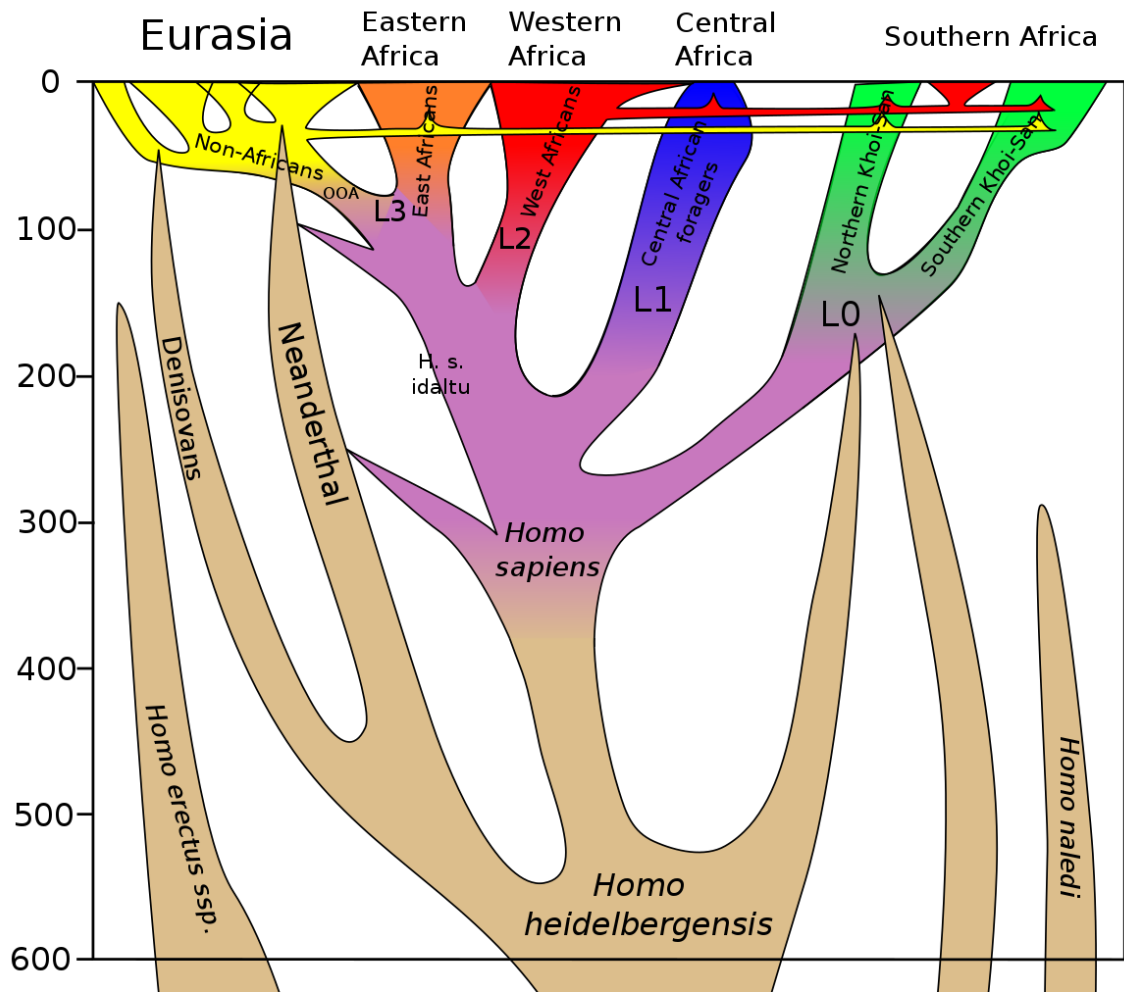
Eurasia

Africa

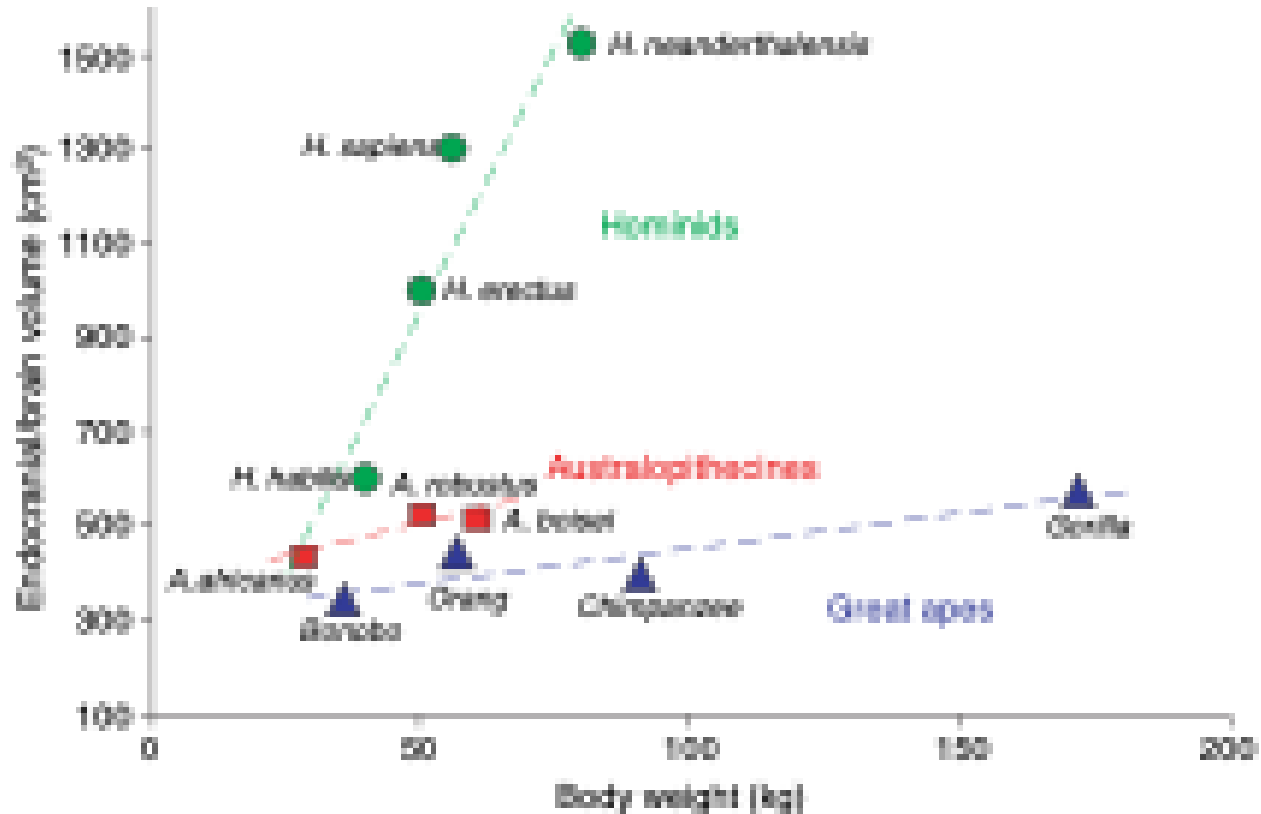


https://en.wikipedia.org/wiki/Homo_erectus

https://commons.wikimedia.org/wiki/File:Homo_lineage_2017update.svg



The evolution of brain size in great apes



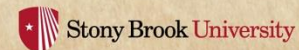
From Roth, G., & Dicke, U. (2005). Evolution of brain and intelligence. *Trends in Cognitive Sciences*, 9, 250-257.

DEVELOPMENT OF STONE TOOL TECHNOLOGY IN EASTERN AFRICA

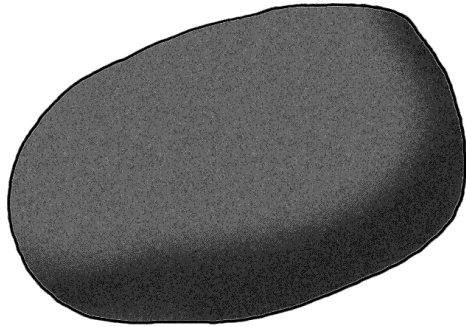
Location of
Lomekwi 3 site



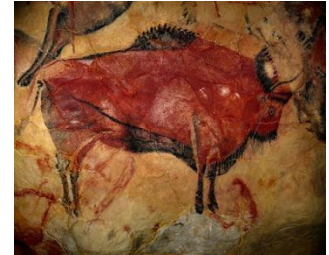
Image Copyrights: Acheulean, Lomekwian tools © MPK-WTAP;
Oldowan tool © Turkana Basin Institute.



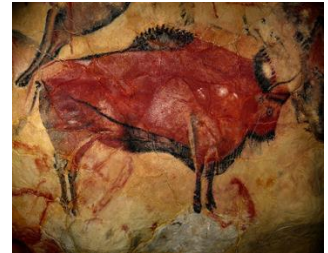
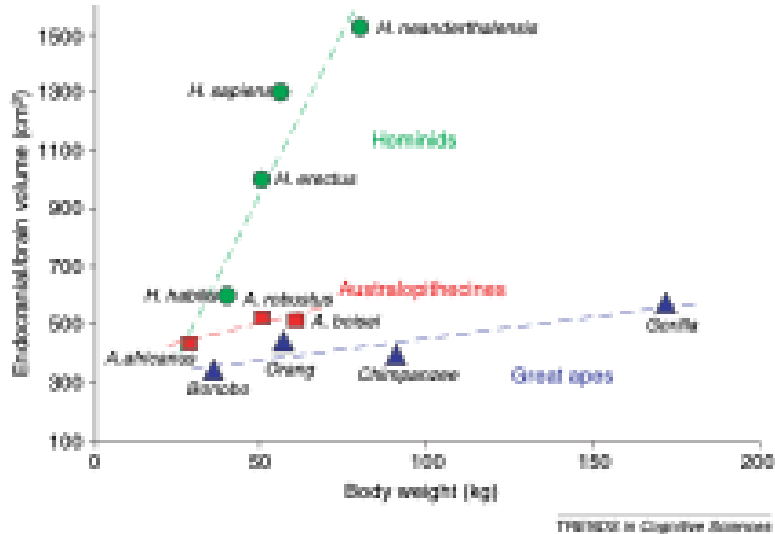
Mousterian
0.3 MYA – 40KYA



Upper Paleolithic Revolution
“Great Leap Forward”, 100-40KYA (?)



Are these informative about language?

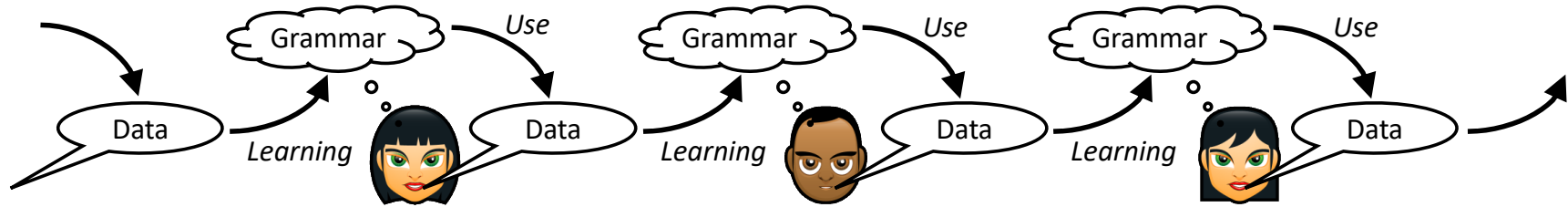


A question from the “ask me anything” box

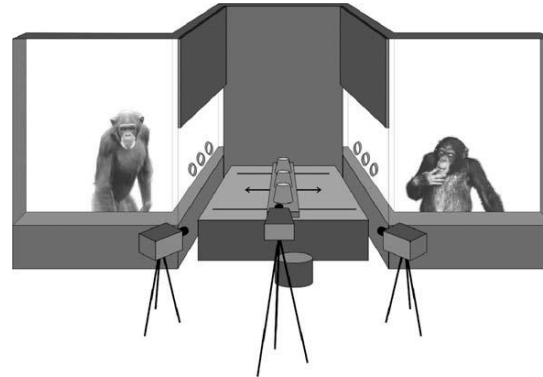
“I have encountered a hypothesis claiming that recursion was a neural ability that originated in tool making which leaked into a pre-existing communication system. I believe this tool making change was the beginning of the Mesolithic. Formerly tools were produced from items which already had some functionality as tools. So improved. Chimpanzees have been seen doing this. But the new approach involved first creating a "core" which had no usefulness as a tool but from which useful tools could be produced. This does seem to be quite a breakthrough. And it seems recursive to me but I can't quite articulate how exactly.

So, what do think of this hypothesis?”

Experiments looking at iterated learning



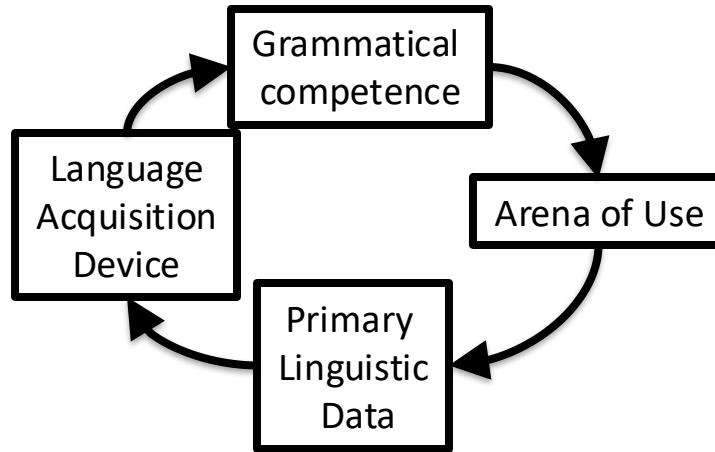
Experiments on learning and cognition in our closest relatives



The core of the argument (from Pinker & Bloom, 1990)

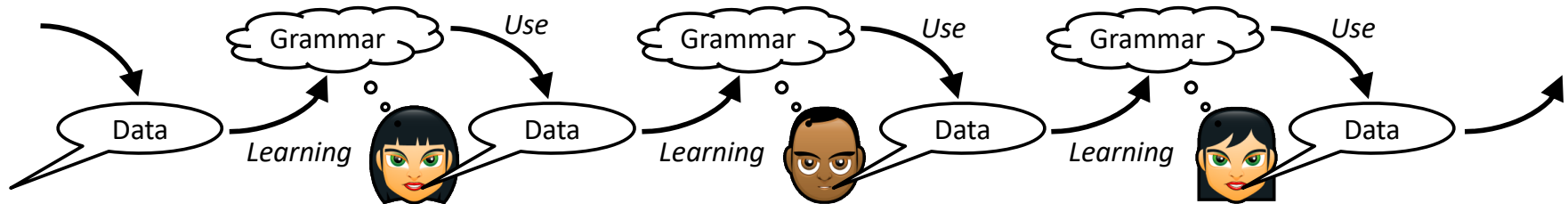


*“All we have argued is that human language, like other specialized **biological** systems, evolved by natural selection. Our conclusion is based on two facts that we would think would be entirely uncontroversial: Language shows signs of complex design for the communication of propositional structures, and the **only explanation** for the origin of organs with complex design is the process of natural selection.”* (p. 726)



Language is transmitted
via repeated **learning**
and **use**

Language is shaped as a
consequence of these
processes



Upper: from Hurford, J. R. (1990). Nativist and functional explanations in language acquisition. In I. M. Roca (Ed.), *Logical issues in language acquisition* (pp. 85–136). Dordrecht: Foris.
Lower: from Smith, K. (2022). How language learning and language use create linguistic structure. *Current Directions in Psychological Science*, 31, 177-186.

Uniformitarianism (in geology)

James Hutton (1726-1797)



**Uniformitarianism: the present
is the key to the past**



John Playfair
(1748-1819)

“On us who saw these phenomenon for the first time the impression will not easily be forgotten...We felt necessarily carried back to a time when the schistus on which we stood was yet at the bottom of the sea, and when the sandstone before us was only beginning to be deposited, in the shape of sand or mud, from the waters of the supercontinent ocean... The mind seemed to grow giddy by looking so far back into the abyss of time; and whilst we listened with earnestness and admiration to the philosopher who was now unfolding to us the order and series of these wonderful events, we became sensible how much further reason may sometimes go than imagination may venture to follow.”



Uniformitarianism in evolutionary linguistics

The present is the key to the past

The more we can explain in terms of processes we can observe in the present day, the happier we should be

- Learning and use explain language change visible in the present and the recent historical record
- Can we explain (some of) language origins in terms of the same processes?

Importantly, uniformity of **process**, not of state: we don't have to say languages have always looked as they do now! (see e.g. Heine & Kuteva, 2002)

Language change

Language change (as attested in the historical record / inferable from synchronic data) is a consequence of:

- Speakers trying to convey meaning efficiently
- Hearers trying to infer speaker meaning
- Language learners (and everyone else) seeking regularities in the linguistic data they encounter

These processes are inherent to the transmission of language via learning and (ostensive-inferential) use

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Ad-hoc extension to meet communicative needs



tick-tock!



axe



corkscrew?



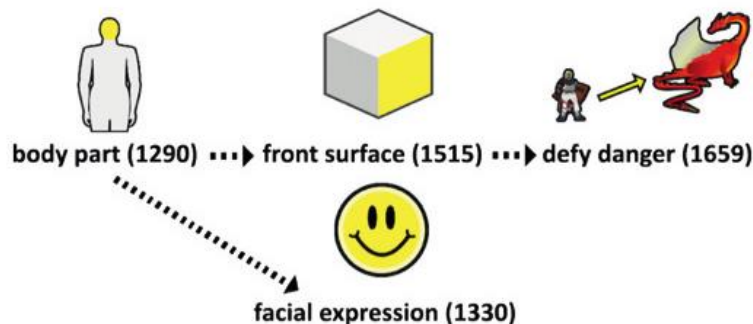
The **beard** is still waiting for his spaghetti



You are **my fire**
The one desire
Believe when I say
I want it that way



“A reef of dead metaphors” (Deutscher, 2005)



From Ramiro, C., Srinivasan, M., Malt, B. C., & Yu, X. (2018). Algorithms in the historical emergence of word senses. *Proceedings of the National Academy of Sciences, USA*, 115, 2323-2328.

“She was *thrilled* to *discover* that the *assessment board* had *decided* to make her *rival redundant*”

thrill: from thirl, “to pierce”

discover: remove the cover from

assessment: from *assidere*, “to sit by” (in judgment)

board: plank

decided: from *de-caedere*, “cut off”

rival: from *rivalis*, someone who shares the same river

redundant: from *redundantem*, “overflow”

From p. 125 of Deutscher, G. (2005). *The Unfolding of Language*. New York, NY: Picador.

Grammaticalization

E.g.: development of future tense markers from verbs of motion

I am going to Gothenburg

MOTION

I am going to stay at home

INTENTION

It is going to rain

FUTURE

Grammaticalization

E.g.: development of future tense markers from verbs of motion

I am going to Gothenburg

MOTION (+ INTENTION)

I am going to stay at home

INTENTION

It is going to rain

FUTURE

Grammaticalization

E.g.: development of future tense markers from verbs of motion

I am going to Gothenburg

MOTION (+ INTENTION)

I am going to buy you a gift!

MOTION + INTENTION

I am going to stay at home

INTENTION

It is going to rain

FUTURE

Grammaticalization

E.g.: development of future tense markers from verbs of motion

I am going to Gothenburg

MOTION (+ INTENTION)

I am going to buy you a gift!

MOTION + INTENTION

I am going to stay at home

INTENTION (+ FUTURE)

It is going to rain

FUTURE

Grammaticalization

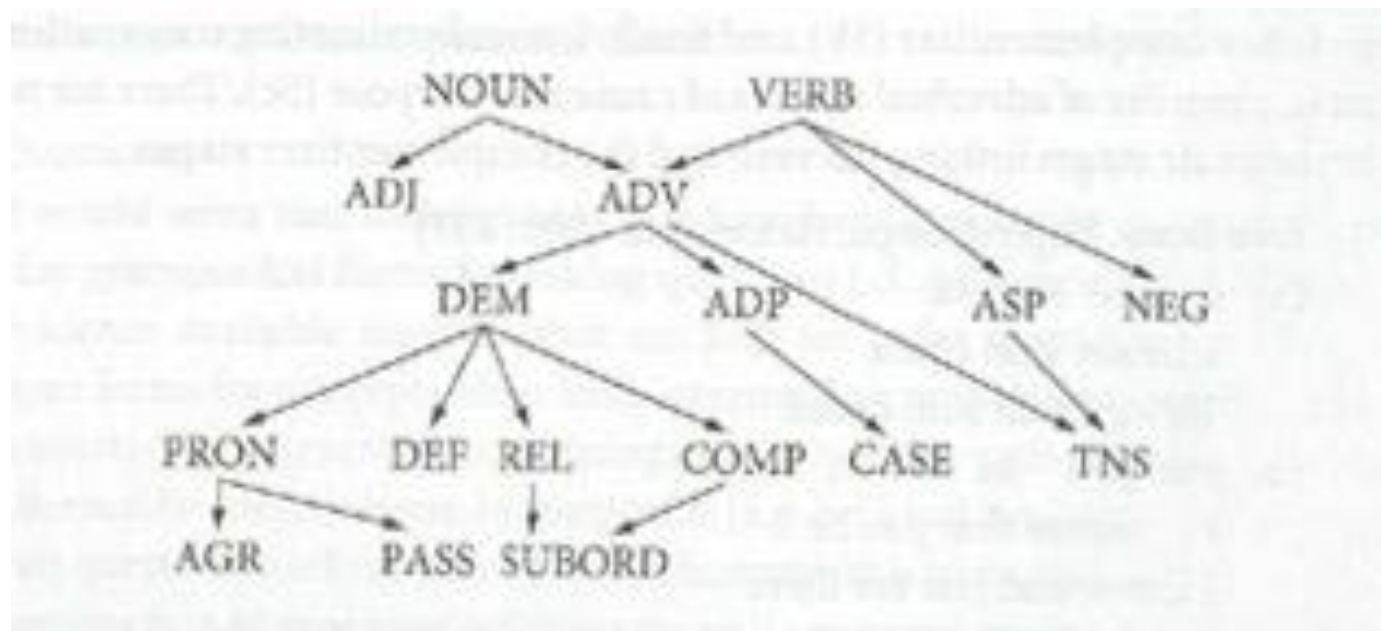
E.g.: development of future tense markers from verbs of motion

<i>I am going to Gothenburg</i>	MOTION (+ INTENTION)
<i>I am going to buy you a gift!</i>	MOTION + INTENTION
<i>I am going to stay at home</i>	INTENTION (+ FUTURE)
<i>I am going to stay at home tomorrow</i>	INTENTION + FUTURE
<i>It is going to rain</i>	FUTURE

Grammaticalization

E.g.: development of future tense markers from verbs of motion

<i>I am going to Gothenburg</i>	MOTION (+ INTENTION)
<i>I am going to buy you a gift!</i>	MOTION + INTENTION
<i>I am going to stay at home</i>	INTENTION (+ FUTURE)
<i>I am going to stay at home tomorrow</i>	INTENTION + FUTURE
<i>It's gonna to rain</i>	FUTURE



From Heine, B., & Kuteva, T. (2002). On the Evolution of Grammatical Forms. In. A. Wray (Ed.) The Transition to Language (pp. 376-397). Oxford: Oxford University Press.

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nature

WORDS ON THE BRINK

The evolution
of language

NUCLEAR
WEAPONS

How to be an
IAEA inspector

ATMOSPHERIC
HUMIDITY

The human touch

GENE SILENCING
Non-toxic RNA
inhibition

NATUREJOBS
Mentoring
awards

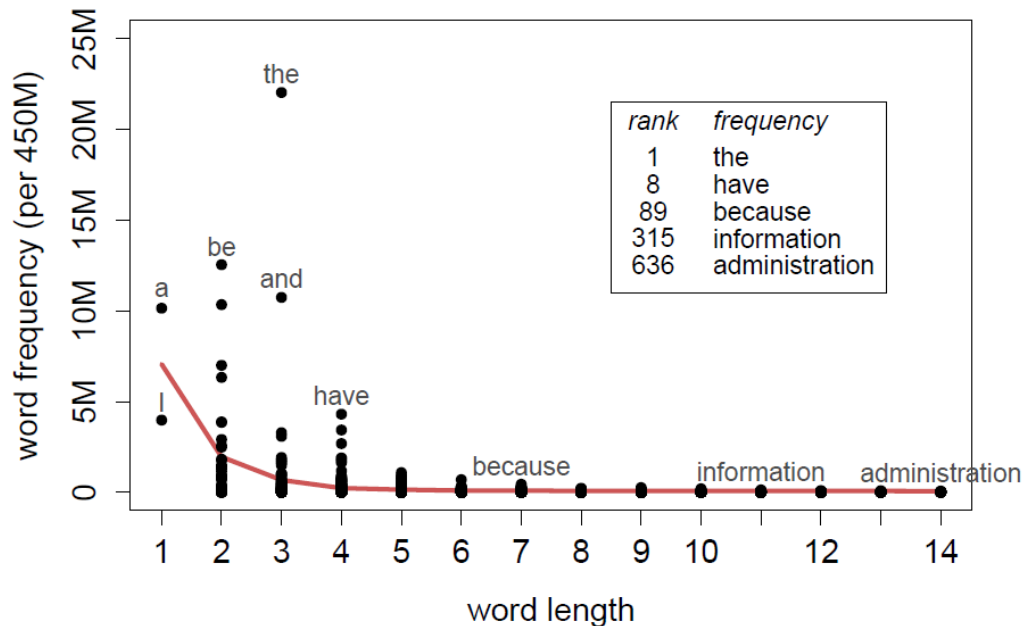


Analological extension & “system pressure”

Cover of issue featuring Lieberman, E., Michel, J. B., Jackson, J.,
Tang, T., & Nowak, M. A. (2007). Quantifying the evolutionary
dynamics of language. *Nature*, 449, 713-716.

Analogical extension & “system pressure”

Frequent words tend to be short (Zipf’s Law of Abbreviation)



From Kanwal, J., Smith, K., Culbertson, J., & Kirby, S. (2017). Zipf’s Law of Abbreviation and the Principle of Least Effort: Language users optimise a miniature lexicon for efficient communication. *Cognition*, 165, 45-52.

Analogical extension & “system pressure”

Frequent words tend to be short (Zipf’s Law of Abbreviation)

But system-level pressures favor **regularity**

English	SG	PL	Percentage of singular
<i>house</i>	49295	9840	83
<i>hare</i>	488	136	78
<i>bear</i>	1182	611	66
<i>window</i>	9936	8506	54
<i>feather</i>	487	810	38
<i>parent</i>	3706	15956	19

From Haspelmath, M. (2014). On system pressure competing with economic motivation. In MacWhinney, B., Malchukov, A., & Moravcsik, E. (Eds) *Competing Motivations in Grammar and Usage* (pp. 197-208). Oxford: Oxford University Press.

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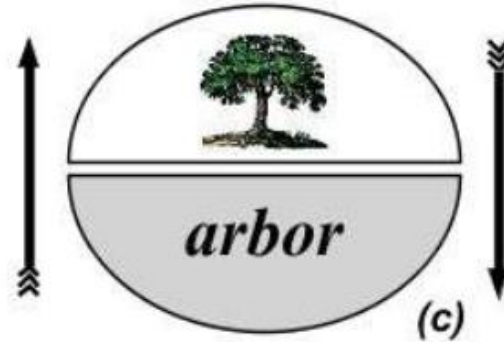
These processes are inherent to the transmission of language via learning and (ostensive-inferential) use

To what extent can these same processes explain the origins of fundamental properties of linguistic systems?

Symbols
Compositionality
Duality of patterning

Where do symbols come from?

- **Icon:** signals resemble meanings
- **Symbol:** *arbitrary* relationship between signal and meaning



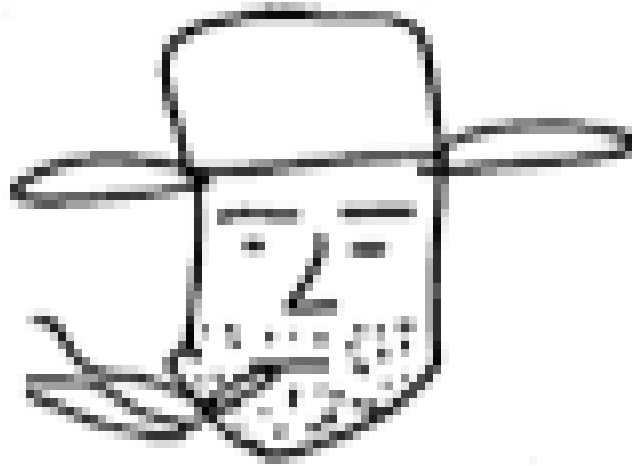
Ritualization in the lab, with humans

Repeated interaction in a Pictionary-like communication task

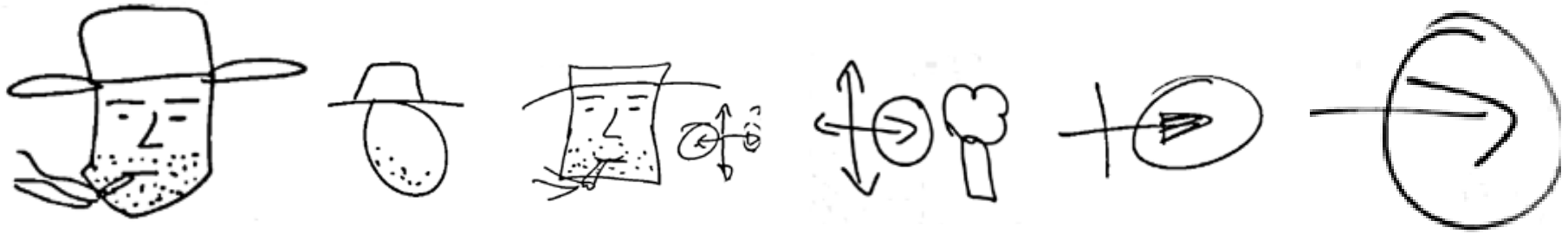


Ritualization in the lab, with humans

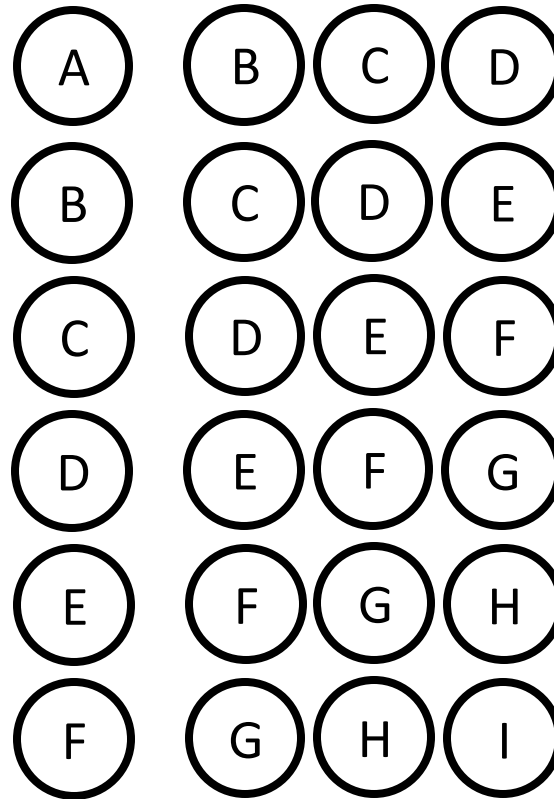
Repeated interaction in a Pictionary-like communication task



Ritualization in the lab



Transmission in laboratory 'societies'



Caldwell, C. A., & Smith, K. (2012). Cultural evolution and the perpetuation of arbitrary communicative conventions in experimental microsocieties. *PLoS ONE*, 7, e43807.

Transmission in laboratory 'societies'

(Round 6)

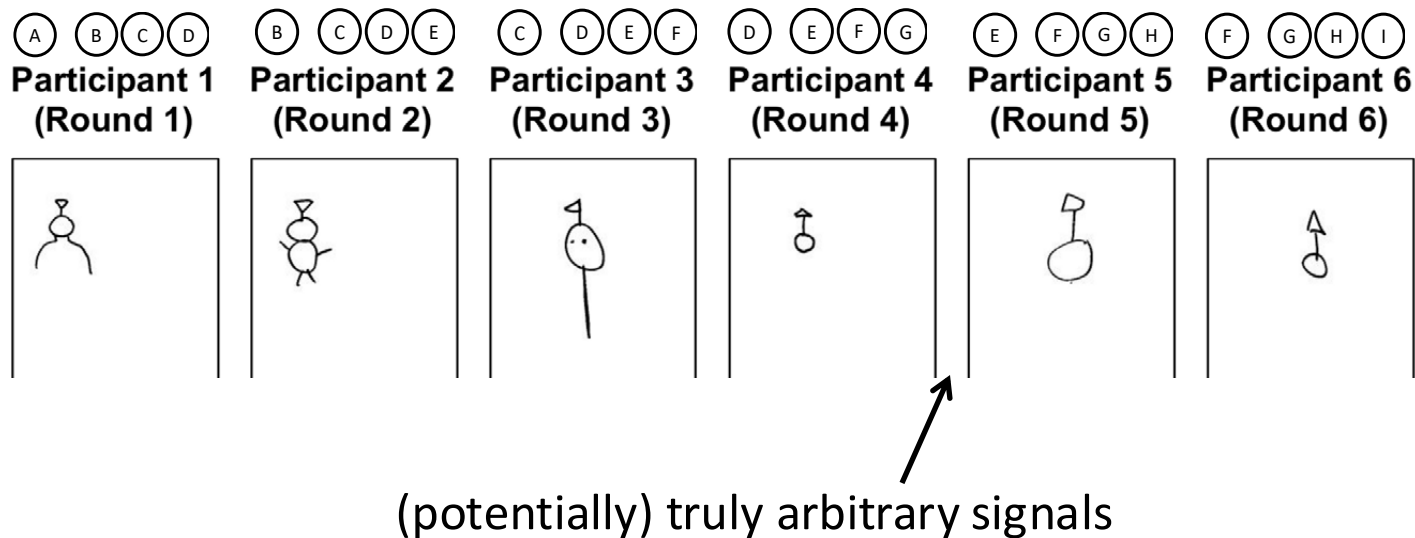


Transmission in laboratory 'societies'

(Rou



Transmission in laboratory 'societies'



How about structure?



Communication is widespread, but language is unique

Language's communicative power comes (in part) from its **structure**

Inventory of meaningless units
(10s)



Inventory of meaningful units
(1000s)



Inventory of meaningful sentences
(∞)

p t d s ɒ k g ɔ ə a ...

ə ðə -əd dɒg kat ðat spɒt ...
(a) (the) (past tense) (dog) (cat) (that) (spot)

the cat spotted the dog a dog spotted the cat
a cat spotted the dog the dog spotted the cat
the cat spotted the cat that spotted a dog ...
the dog spotted the cat that spotted the dog

Language's communicative power comes (in part) from its **structure**

Inventory of meaningless units
(10s)

p t d s ɒ k g ɔ ə a ...

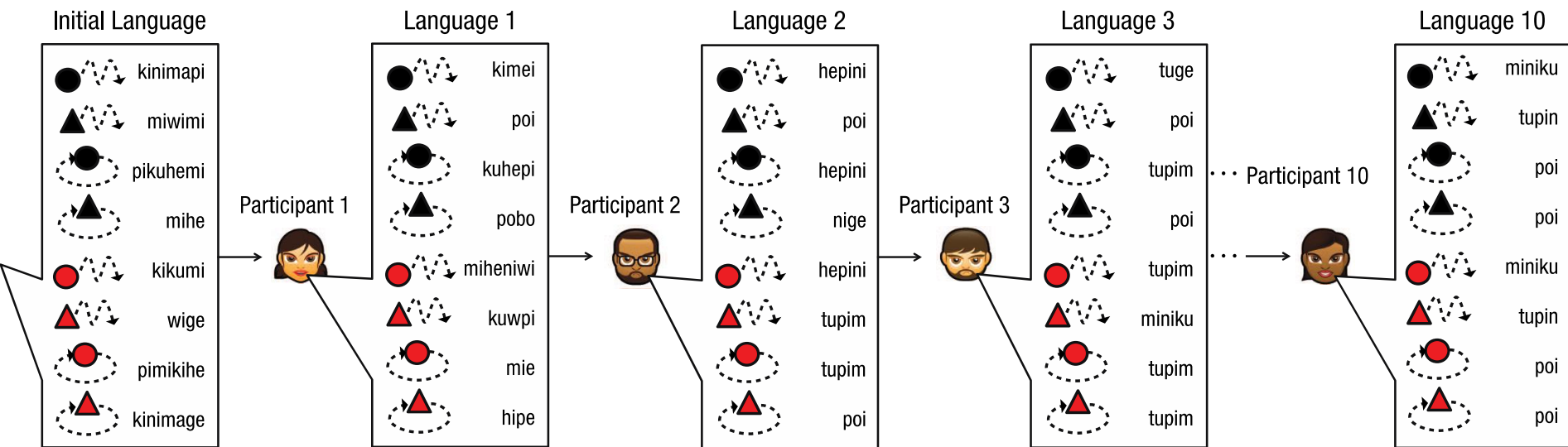
Inventory of meaningful units
(1000s)

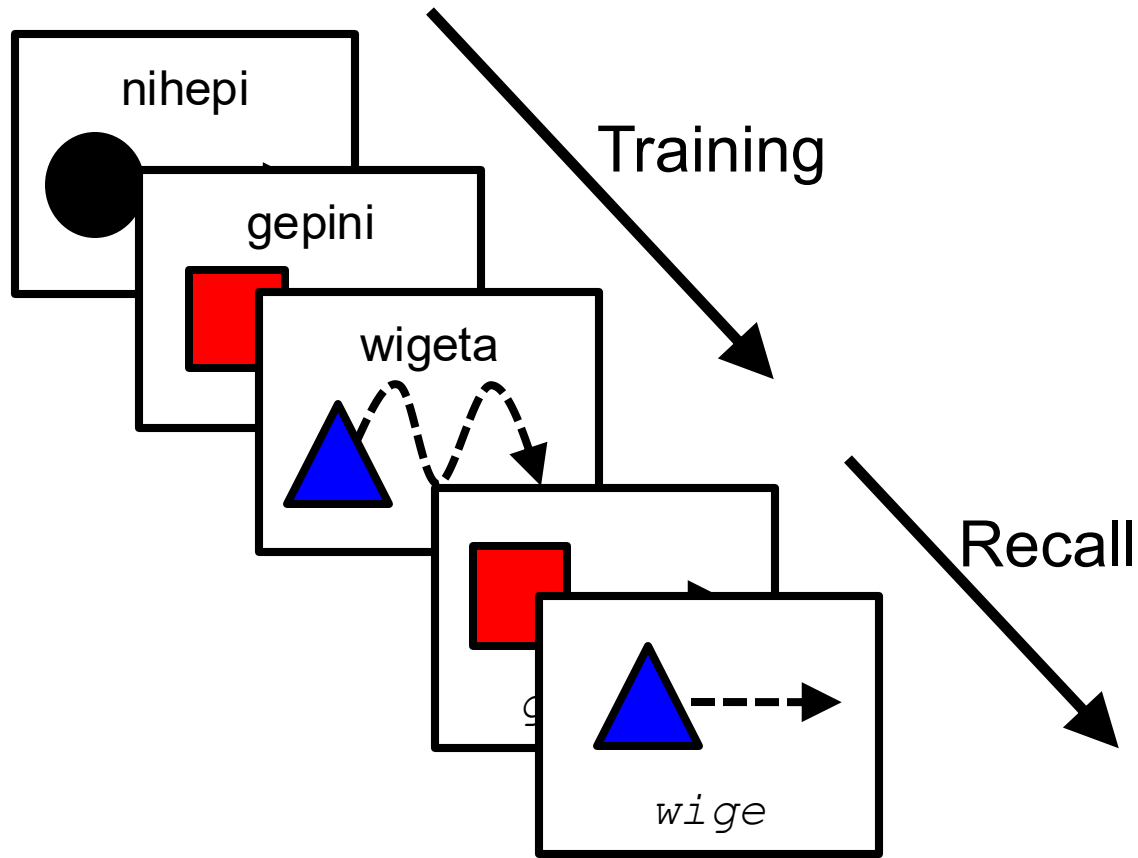
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Inventory of meaningful sentences
(∞)

the cat spotted the dog a dog spotted the cat
a cat spotted the dog the dog spotted the cat
the cat spotted the cat that spotted a dog ...
the dog spotted the cat that spotted the dog

Iterated Learning





Kirby, S., Cornish, H., & Smith, K. (2008). Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language. *PNAS*, 105, 10681-10686.

An initial non-compositional (**holistic**) language from chain 4

--→	wimaku	miniki	gepinini	□
	nihepi	wigemi	mahekuki	○
	wikima	nipikuge	hema	△
↻	miwiniku	pinipi	kihemiwi	□
	kinimapi	wikuki	kikumi	○
	miwimi	nipi	wige	△
↻	gepihemi	kunige	miki	□
	pikuhemi	kimaki	pimikihe	○
	mihe	winige	kinimage	△

Seen vs unseen

→			gepinini	□
			mahekuki	○
	wikima	nipikuge		△
↻	miwiniku		kihemiwi	□
		wikuki		○
	miwimi		wige	△
↻	gepihemi			□
	pikuhemi	kimaki		○
		winige	kinimage	△

Generation 0 language from chain 4

-->	wimaku	miniki	gepinini	□
	nihepi	wigemi	mahekuki	○
	wikima	nipikuge	hema	△
↻	miwiniku	pinipi	kihemiwi	□
	kinimapi	wikuki	kikumi	○
	miwimi	nipi	wige	△
↻	gepihemi	kunige	miki	□
	pikuhemi	kimaki	pimikihe	○
	mihe	winige	kinimage	△

Generation 1 language from chain 4

--→	nige	miniku	poh	□
	mip	mpo	miniku	○
	tuge	tuge	weg	△
↻	pemini	kupini	pon	□
	kimei	miwn	miheniw	○
	poi	mhip	kuwpi	△
↻	hepinimi	himini	hipe	□
	kuhepi	wige	mie	○
	pobo	tupim	hipe	△

Generation 2 language from chain 4

-->	nige	tuge	tuge	□
	nige	nige	tuge	○
	nige	tuge	mpo	△
↗↘	nige	tuge	mihenu	□
	hepini	miniku	hepini	○
	poi	mpo	tupim	△
↻	hepini	miniku	tupim	□
	hepini	tupim	tupim	○
	nige	tupim	poi	△

Generation 3 language from chain 4

-->	mihenu	tuge	tuge	□
	tuge	tuge	tuge	○
	nige	tuge	tuge	△
↗↘	miniku	tuge	minihu	□
	tuge	tupim	tupim	○
	poi	tuge	miniku	△
↻	mihenu	tupim	tupim	□
	tupim	tupim	tupim	○
	poi	tupim	tupim	△

Generation 4 language from chain 4

-->	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
		tuge	tuge	△
↗↘	mihunu	tupim	miniku	□
	tupim	tupim	tupim	○
	poi	miniku	miniku	△
↻	tupim	tupim	tupim	□
	tupim	tupim	tupim	○
	poi	tupim	tupim	△

Generation 5 language from chain 4

-->	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
↗↘	minuhu	tupim	tupim	□
	tupim	tupim	tupim	○
	miniku	tupim	miniku	△
↻	tupim	tupim	tupim	□
	tupim	tupim	tupim	○
	poi	tupim	tupim	△

Generation 6 language from chain 4

--→	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
↗↘→	miniku	tupin	tupim	□
	miniku	miniku	miniku	○
	miniku	tupin	tupin	△
↻	poi	tupin	tupim	□
	poi	poi	poi	○
	poi	tupin	tupim	△

Generation 7 language from chain 4

--→	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
↻	miniku	miniku	tupim	□
	miniku	miniku	miniku	○
	miniku	tupin	miniku	△
↻	poi	poi	tupim	□
	poi	poi	poi	○
	poi	tupin	poi	△

Generation 8 language from chain 4

--→	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
↻	tupim	tupim	tupim	□
	miniku	miniku	miniku	○
	tupin	tupin	tupin	△
↻	poi	poi	poi	□
	poi	poi	poi	○
	poi	poi	poi	△

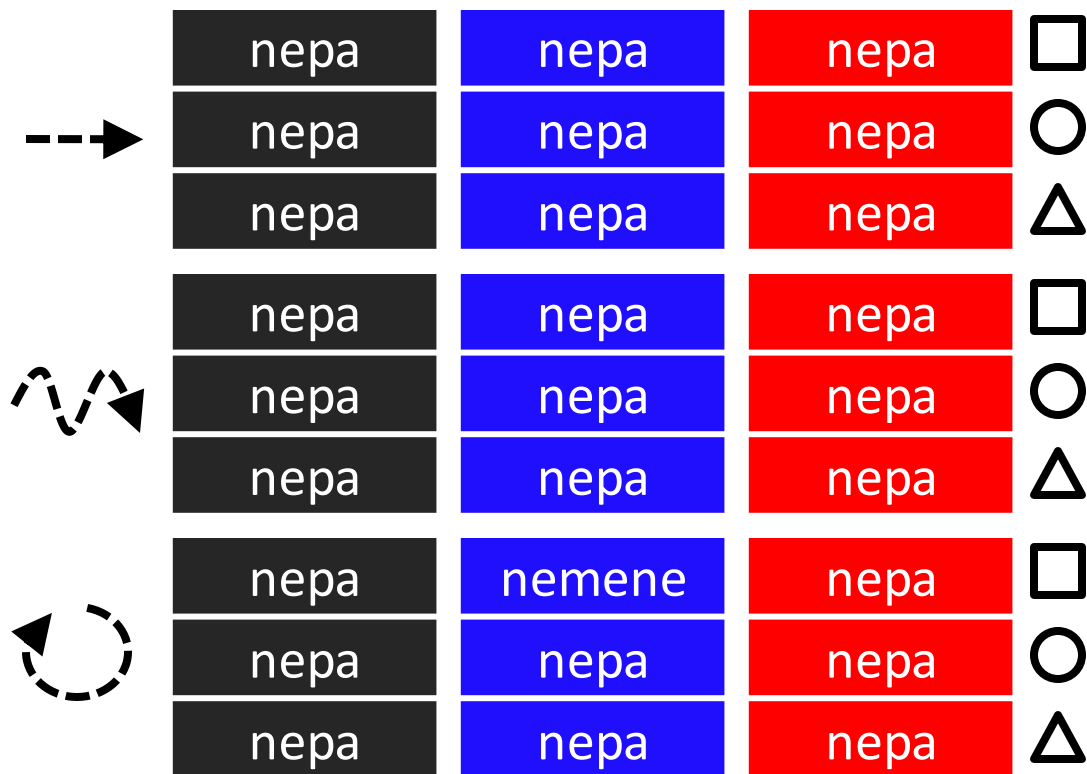
Generation 9 language from chain 4

--→	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
↻	tupim	tupim	tupim	□
	miniku	miniku	miniku	○
	tupin	tupin	tupin	△
↻	poi	poi	poi	□
	poi	poi	poi	○
	poi	poi	poi	△

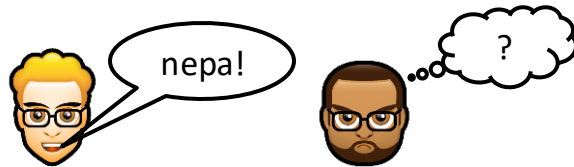
Generation 10 language from chain 4

-->	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
~>	tupim	tupim	tupim	□
	miniku	miniku	miniku	○
	tupin	tupin	tupin	△
↻	poi	poi	poi	□
	poi	poi	poi	○
	poi	poi	poi	△

Final language from chain 1 (!)



The languages become **degenerate**



Learnability and degeneracy

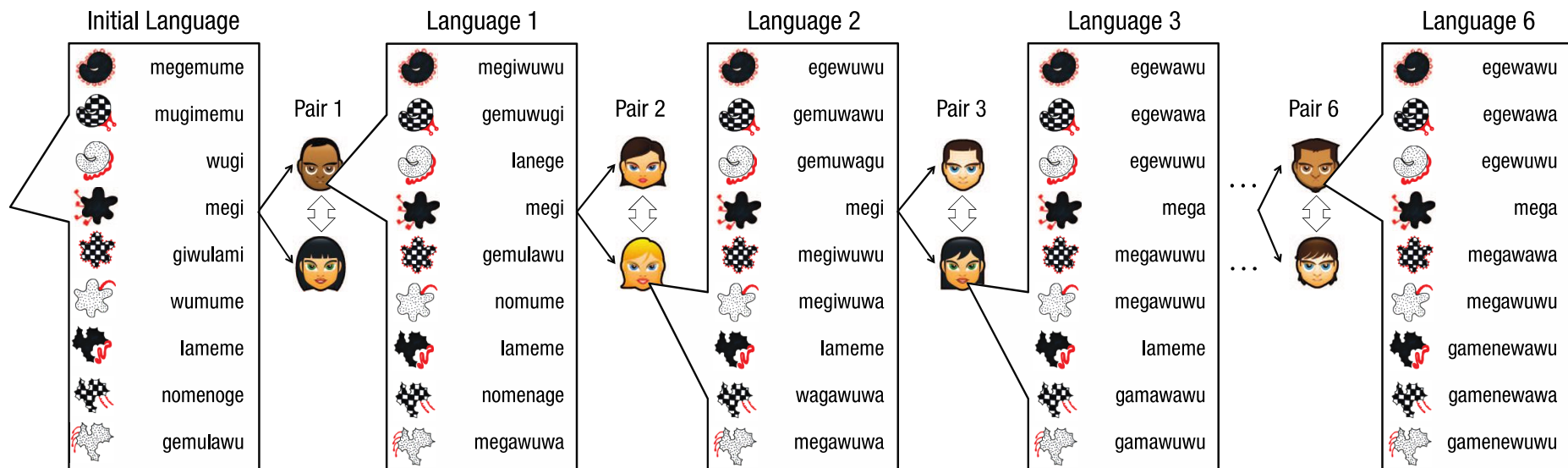
Learners prefer simpler languages

The only pressure in Kirby, Cornish & Smith (2008) Experiment 1 is **learnability**

- The languages don't need to be **expressive**
- They get very simple

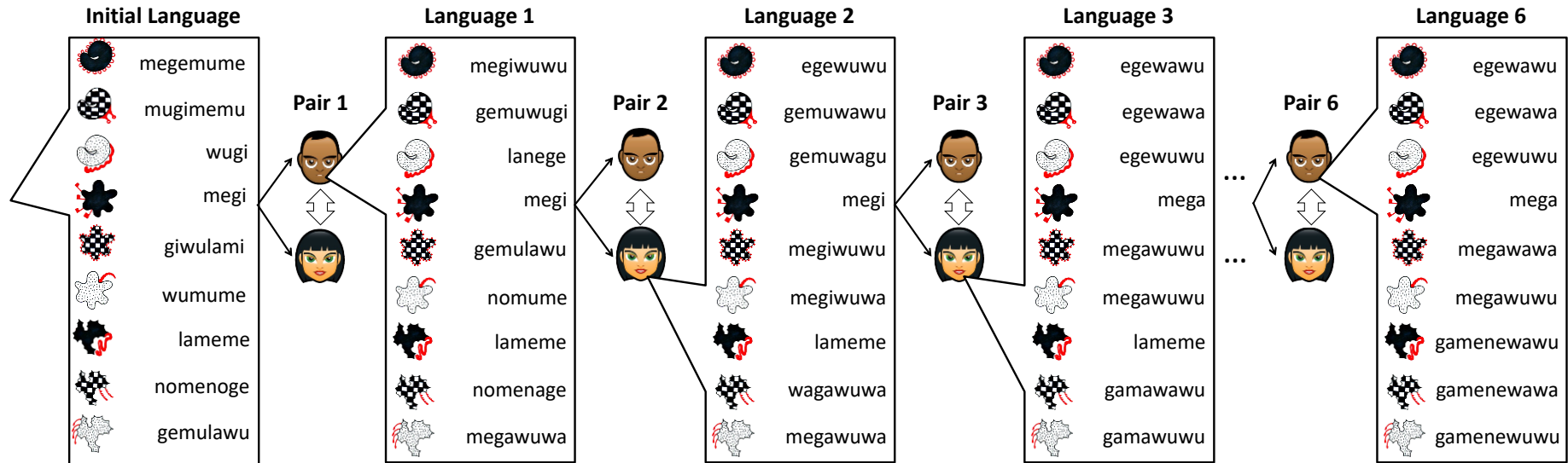
Can we add in a pressure for expressivity?

Kirby, Tamariz, Cornish & Smith (2015): Adding communication



Kirby, S., Tamariz, M., Cornish, H., & Smith, K. (2015). Compression and Communication in the Cultural Evolution of Linguistic Structure. *Cognition*, 141, 87-102.

Kirby, Tamariz, Cornish & Smith (2015): Adding communication, removing learning



An initial non-compositional mapping

 megemume	 megi	 lameme
 mugimemu	 giwulami	 nomenoge
 wugi	 wumume	 gemulawu
 lamege	 wulamugi	 megiwuwa

Chain, Generation 1

 megiwuwu	 megi	 lameme
 gemuwugi	 gemulawu	 nomenage
 lanege	 nomume	 megawuwa
 nomege	 wulagi	 wagawu

Chain, Generation 2

 egewuwu	 megi	 lameme
 gemuwawu	 megiwuwu	 wagawuwa
 gemuwagu	 megiwuwa	 megawuwa
 ege	 wulagi	 wagawuwa

Chain, Generation 3

 egewawu	 mega	 lameme
 egewawa	 megawuwu	 gamawawu
 egewuwu	 megawuwu	 gamawuwu
 ege	 wulagi	 gamene

Chain, Generation 4

 egewawu	 mega	 gamenewawu
 egewawa	 megawawa	 gamenewawa
 egewuwu	 megawuwu	 gamenewuwu
 ege	 wulagi	 gamane

Chain, Generation 5

 egewawu	 mega	 gamenewawu
 egewawa	 megawawa	 gamenewawa
 egewuwu	 megawuwu	 gamenewuwu
 ege	 wulagi	 gamane

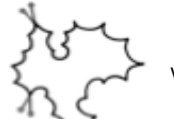
Chain, Generation 6

 egewawu	 mega	 gamenewawu
 egewawa	 megawawa	 gamenewawa
 egewuwu	 megawuwu	 gamenewuwu
 ege	 wulagi	 gamane

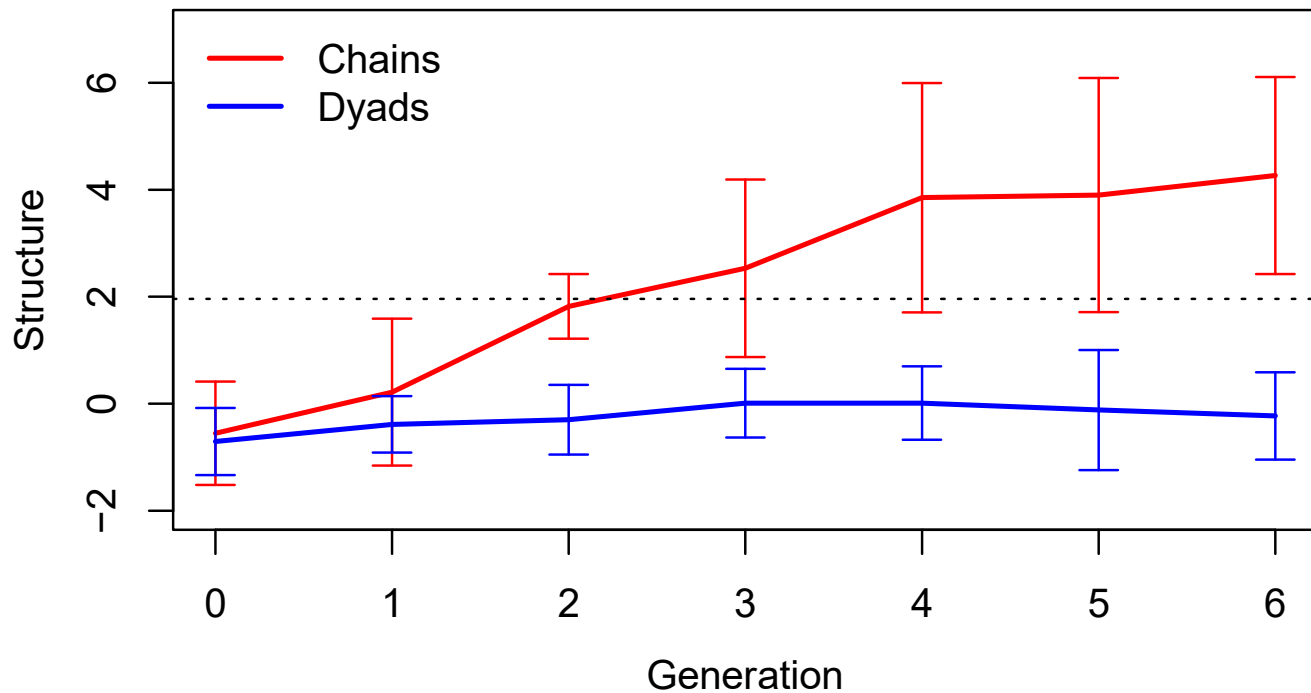
Chain, Generation 6

 egewawu	 mega	 gamenewawu
 egewawa	 megawawa	 gamenewawa
 egewuwu	 megawuwu	 gamenewuwu
 ege	 wulagi	 gamane

A final non-compositional language from a dyad

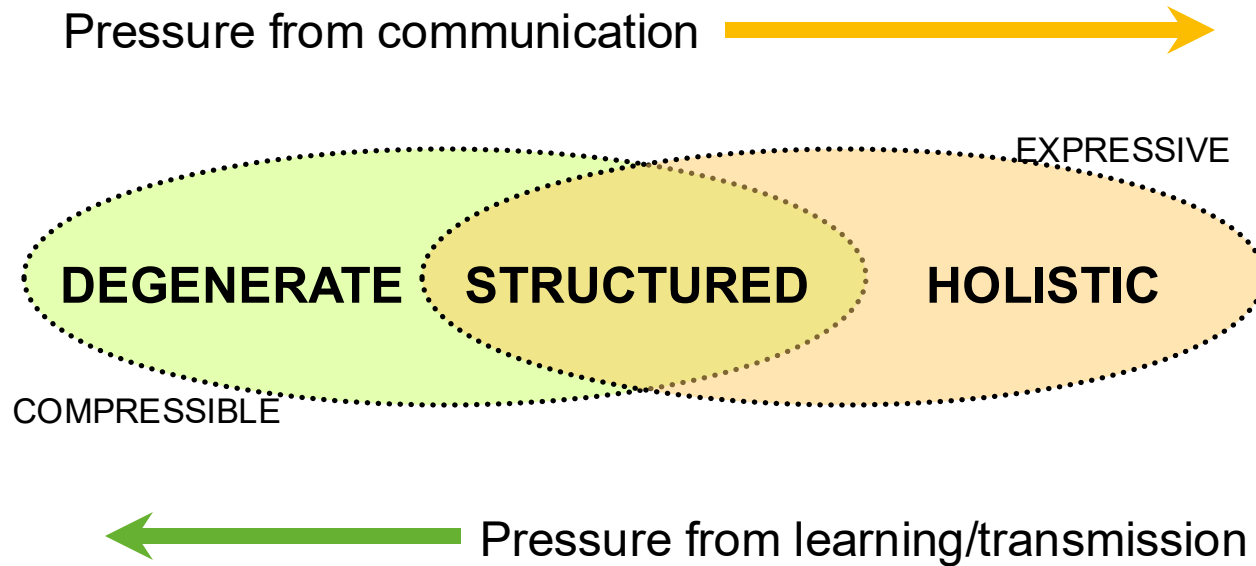
 manunumoko	 moko	 konu
 wekihumanunu	 mokowekihu	 lawa
 makihu	 mahiku	 wekihulawa
 manunumonu	 nomu	 wekihu

Structure in chains, not dyads

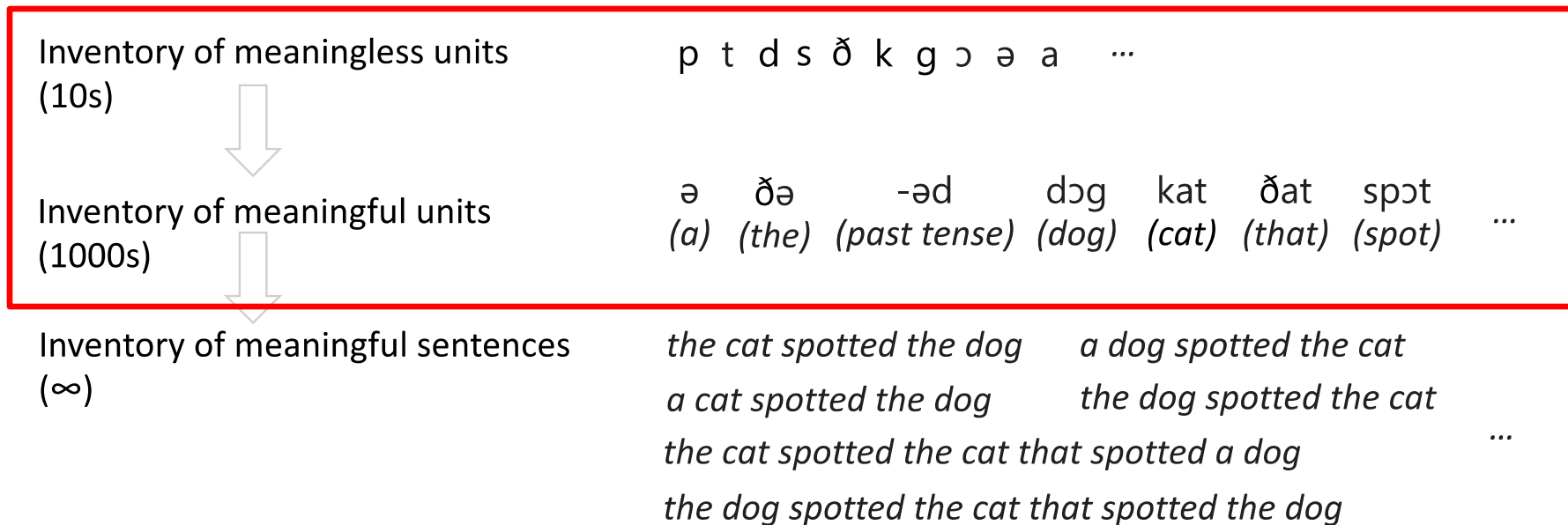


Learnability + expressivity = **structure**

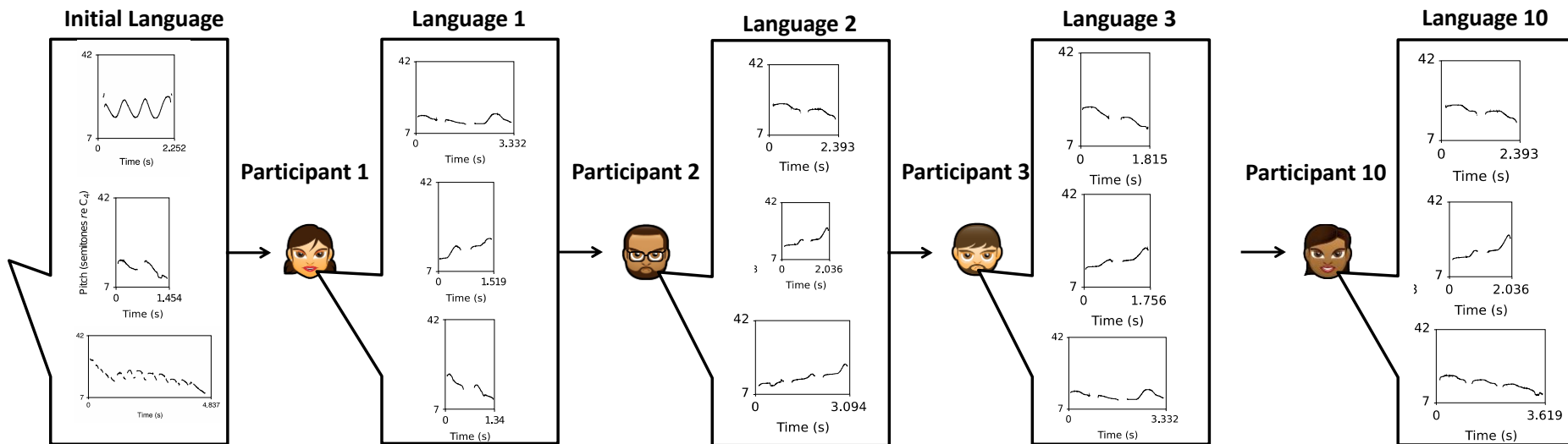
Structure as a trade-off between compressibility and expressivity that plays out over cultural transmission



Language's communicative power comes (in part) from its **structure**

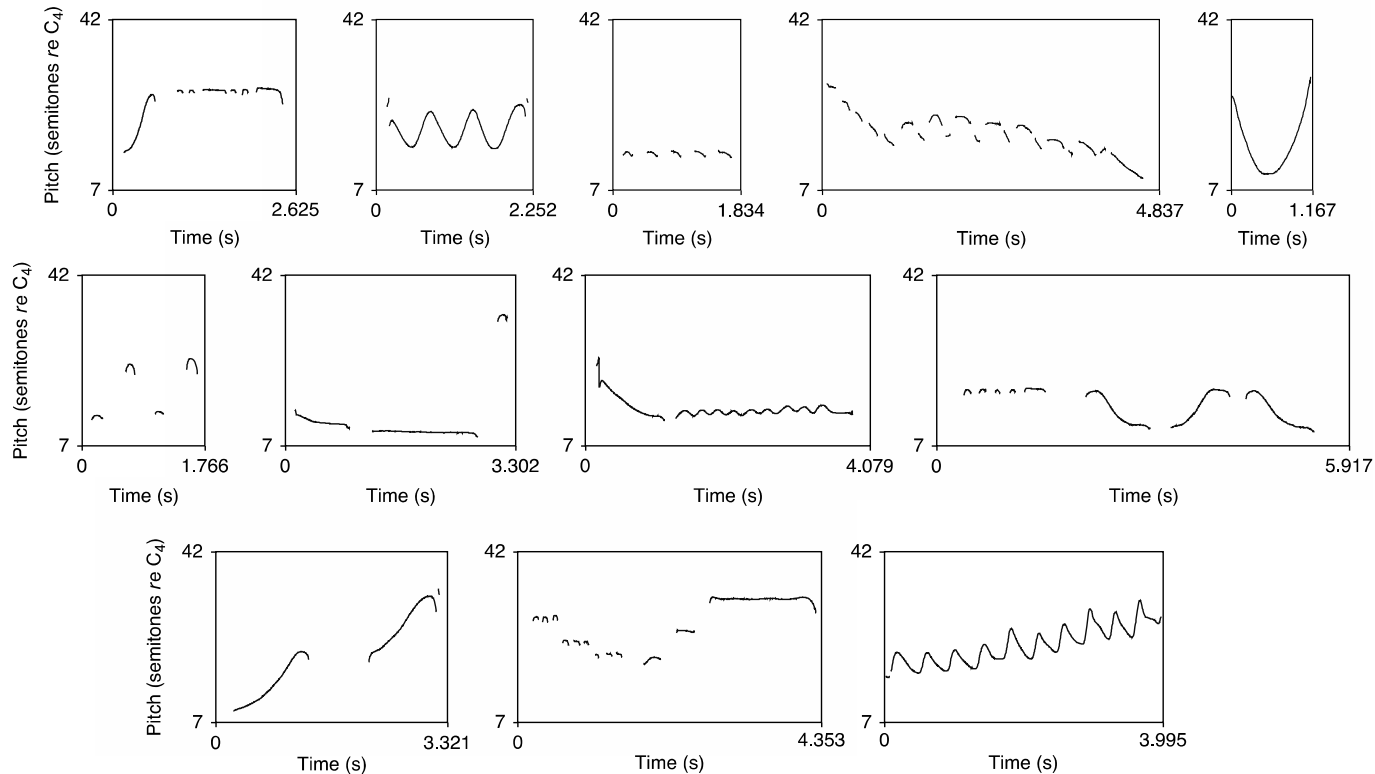


Iterated Learning with whistles

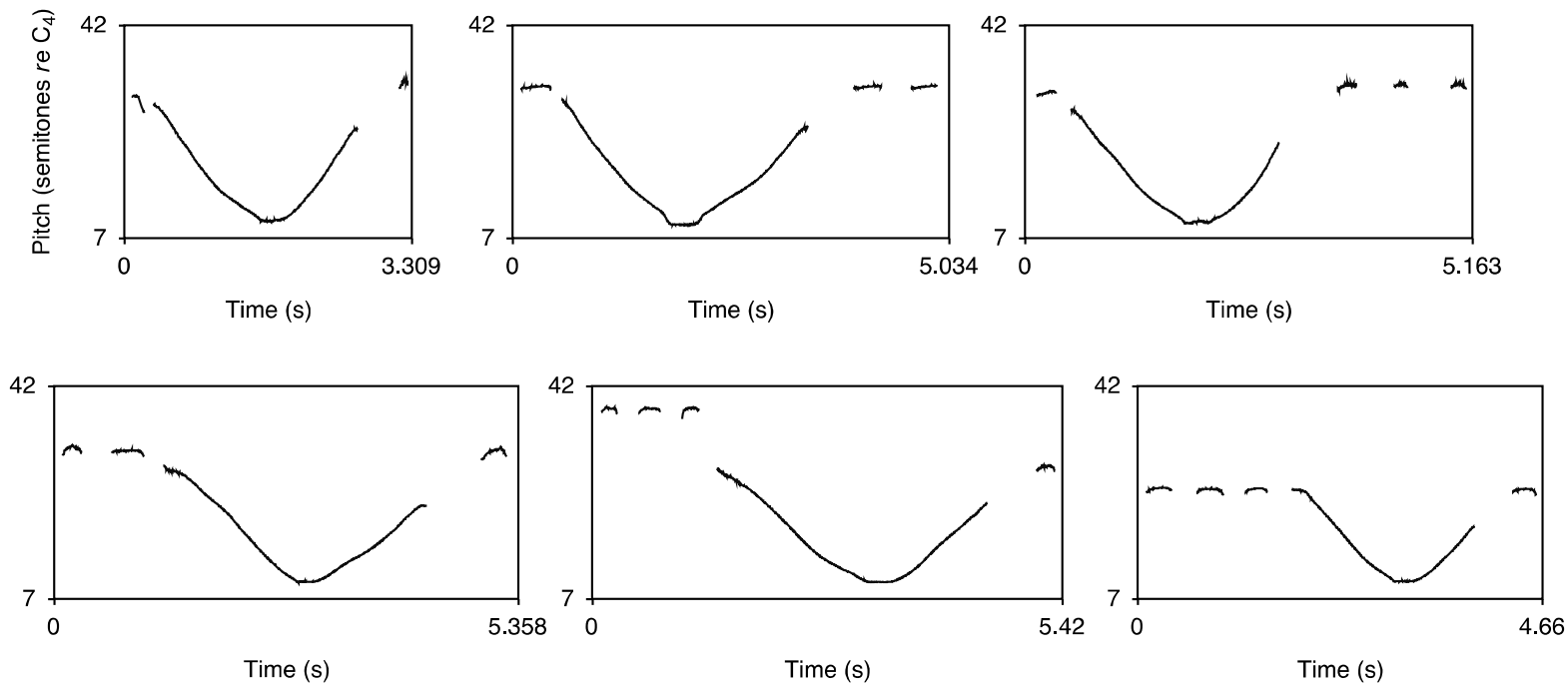


Verhoef, T., Kirby, S., & de Boer, B. (2014). Emergence of combinatorial structure and economy through iterated learning with continuous acoustic signals. *Journal of Phonetics*, 43,

The initial whistle set



(Part of) A generation 10 whistle set



The core of the argument (from Pinker & Bloom, 1990)



*“All we have argued is that human language, like other specialized **biological** systems, evolved by natural selection. Our conclusion is based on two facts that we would think would be entirely uncontroversial: Language shows signs of complex design for the communication of propositional structures, and the **only explanation** for the origin of organs with complex design is the process of natural selection.”* (p. 726)

Next class: iterated learning advanced topics

- Fine-grained adaptation to different communicative and learning pressures
- Iterated learning in non-humans
- **Cognitive prerequisites for iterated learning**